Engineering, inventions and design

1. Read the article once for meaning. 2. Read it again and look at the **highlighted new words** and the **blue phrases** in context. 3. Study the tables. 4. Use at least 5 new words in speaking or writing this week.

Article: The art of making things (178 words)

Every **invention** begins as an idea on paper. An engineer drafts a **blueprint**, builds a **prototype** and tests it repeatedly before approving a final design. Complex machines depend on a precise **mechanism**, and each **component** must fit exactly. Large products are made by **manufacture** on an **assembly line**, where **robotics** has replaced much manual work and greatly improved **efficiency**. Designers choose a **durable** material when they want a product to last.

Civil engineering shapes the world around us. A tall **structure** needs deep **foundation** work; a **bridge** must carry heavy loads, and a **tunnel** may cross mountains or rivers. Builders **reinforce** concrete with steel to make it stronger. The energy sector has seen its own changes: a modern wind **turbine** produces clean **renewable energy**, a **fuel cell** powers a vehicle with no exhaust, and a safe **circuit** is vital in every device.

Each **innovation** builds on earlier ones, and engineers often **go back to the drawing board** when a test fails. Success depends on careful testing as much as on creativity, because a small flaw can **spoil the whole project**.

Useful phrases from the article

PHRASE	MEANING
go back to the drawing board	start the design again
spoil the whole project	ruin everything

The 20 new words

WORD		MEANING	EXAMPLE
invention	n	a new thing created by someone	*The invention of the printing press changed the world.*
prototype	n	the first model of a product	*The prototype was tested for a year.*
mechanism	n	the parts of a machine that work together	*The clock's mechanism is very complex.*
efficiency	n	doing something with little waste	*Better insulation improves energy efficiency.*
durable	adj	strong and lasting a long time	*Durable materials reduce waste.*
renewable energy	n	energy from sources that do not run out	*Wind is a renewable energy source.*
turbine	n	a machine that turns movement into power	*The wind turbine produces electricity.*

circuit	n	a path for electricity	*The circuit failed during the test.*
blueprint	n	a detailed plan of a design	*The blueprint shows every room.*
manufacture	v	to make goods in large numbers	*The company manufactures car parts.*
assembly line	n	a line of workers or machines making a product	*Robots work on the assembly line.*
component	n	one part of a larger machine	*A faulty component caused the fault.*
structure	n	a building or the way parts are arranged	*The structure can resist earthquakes.*
bridge	n	a structure that crosses a river	*The bridge took five years to build.*
tunnel	n	an underground passage	*The tunnel connects two countries.*
foundation	n	the base of a building	*Deep foundations keep the tower stable.*
reinforce	v	to make stronger	*Steel bars reinforce the concrete.*
fuel cell	n	a device that makes electricity from chemicals	*Fuel cells power some buses.*
robotics	n	the science of building robots	*Robotics is used in surgery.*
innovation	n	a new method or device	*Innovation in batteries is rapid.*

Revision: 10 words from last week

WORD		MEANING
diagnosis	n	identifying an illness
therapy	n	treatment of a disease
surgery	n	medical treatment by cutting the body
antibiotic	n	a medicine that kills bacteria
infection	n	a disease caused by germs
pandemic	n	a disease that spreads across the world
clinical trial	n	a test of a new medicine on people
dose	n	the amount of medicine taken at one time
prescription	n	a doctor's written order for medicine
recovery	n	getting well again

Practice: fill the gaps

bridge · robotics · foundation · efficiency · structure · reinforce · turbine · durable

1. The _____ took five years to build.

2. _____ is used in surgery.
3. Deep _____ keep the tower stable.
4. Better insulation improves energy _____.
5. The _____ can resist earthquakes.
6. Steel bars _____ the concrete.
7. The wind _____ produces electricity.
8. _____ materials reduce waste.

More words on this topic (not in the article: choose the 10 most useful for you)

alloy (n): a mixture of metals
battery (n): a device that stores power
cable (n): a thick wire
capacity (n): the amount something can hold
chemical (n): a substance made by chemistry
concrete (n): a hard building material
conductor (n): a material that carries electricity
construction (n): the building of something
crane (n): a machine for lifting
demolish (v): to knock down
device (n): a tool or machine
dismantle (v): to take apart
dynamic (adj): full of energy and change
electricity (n): power from electric charge
engine (n): a machine that turns fuel into movement
equipment (n): tools for a job
fault (n): a flaw
generator (n): a machine that makes electricity
hydroelectric (adj): using water to make power
inventor (n): a person who creates something new
machinery (n): machines as a group
metal (n): a hard material such as iron
nuclear (adj): connected with atoms
pipeline (n): a long pipe for oil or gas
plastic (n): a light man-made material
power plant (n): a place where electricity is made
sensor (n): a device that detects changes
solar (adj): from the sun
steel (n): a strong metal made from iron
technician (n): a skilled worker

Answers

1. bridge (bridge)
2. Robotics (robotics)
3. foundations (foundation)
4. efficiency (efficiency)
5. structure (structure)
6. reinforce (reinforce)
7. turbine (turbine)
8. Durable (durable)